DSA0601- DATA HANDLING AND VISUALIZATION

EXPERIMENTS 6 – 10

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EXPERIMENT 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

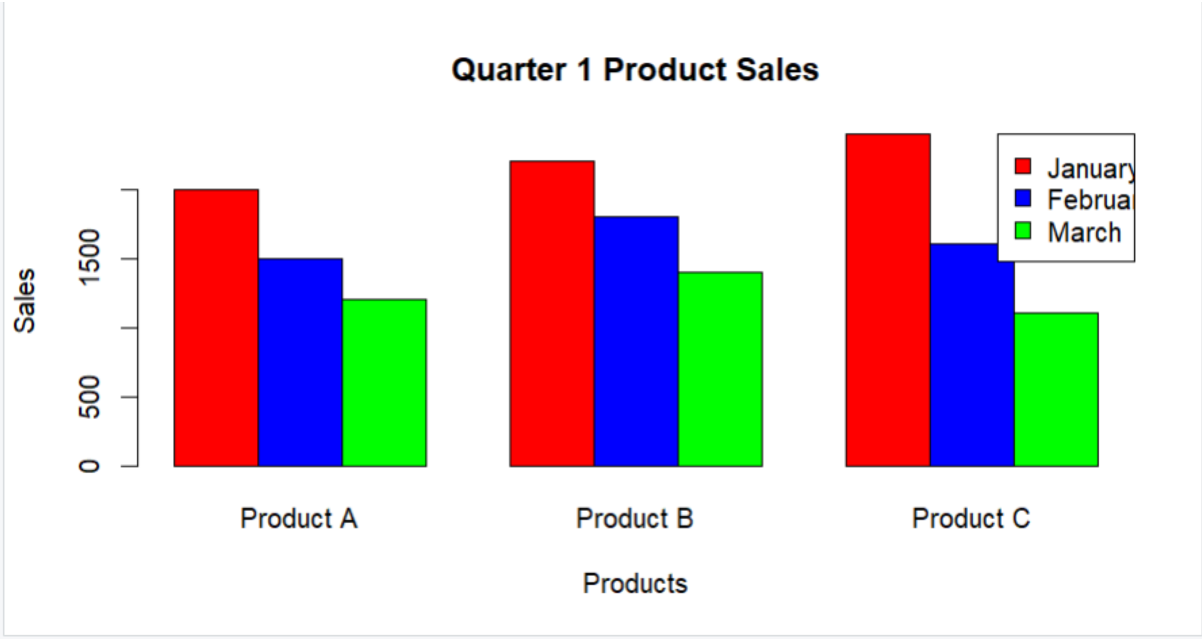
CODE:

```
# Dataset
product <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
# 1. Grouped Bar Chart
sales <- as.matrix(product[,2:4])
barplot(sales,
        beside=TRUE,
        col=c("red","blue","green"),
```

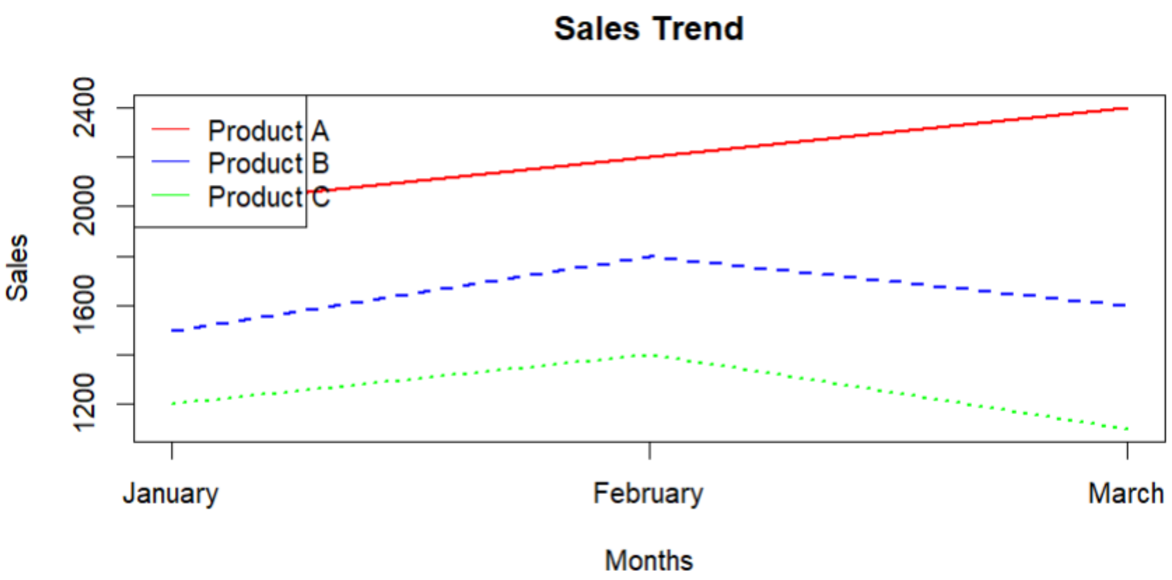
```
        names.arg=product$Product,
        xlab="Products",
        ylab="Sales",
        main="Quarter 1 Product Sales")
legend("topright",
       legend=c("January","February","March"),
       fill=c("red","blue","green"))
# 2. Stacked Area Chart
months <- c(1,2,3)
matplot(months, t(sales),
        type="l",
        lwd=2,
        col=c("red","blue","green"),
        xaxt="n",
        xlab="Months",
        ylab="Sales",
        main="Sales Trend")
axis(1, at=1:3,
     labels=c("January","February","March"))
legend("topleft",
       legend=product$Product,
       col=c("red","blue","green"),
       lty=1)
# 3. Table
Product
```

QUESTIONS:

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3. Build a table to show the monthly sales data for each product. Label the table elements.

```
#-----------------------------
# 3. Table
#-----------------------------
product
  Product January February March
Product A    2000     2200  2400
Product B    1500     1800  1600
Product C    1200     1400  1100
```

EXPERIMENT 7

Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in $)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

CODE:

```
# Customer Demographics
customer <- data.frame(
 CustomerID=c(1,2,3),
 Age=c(28,35,42),
 Gender=c("Female","Male","Female"),
 Income=c(50000,60000,75000)
)
# 1. Age Distribution Bar Chart
barplot(customer$Age,
    names.arg=customer$CustomerID,
    col="skyblue",
    xlab="Customer ID",
    ylab="Age",
    main="Customer Age Distribution")
# 2. Gender Pie Chart
gender_count <- table(customer$Gender)
```

pie(gender_count,

col=c("pink","lightblue"),

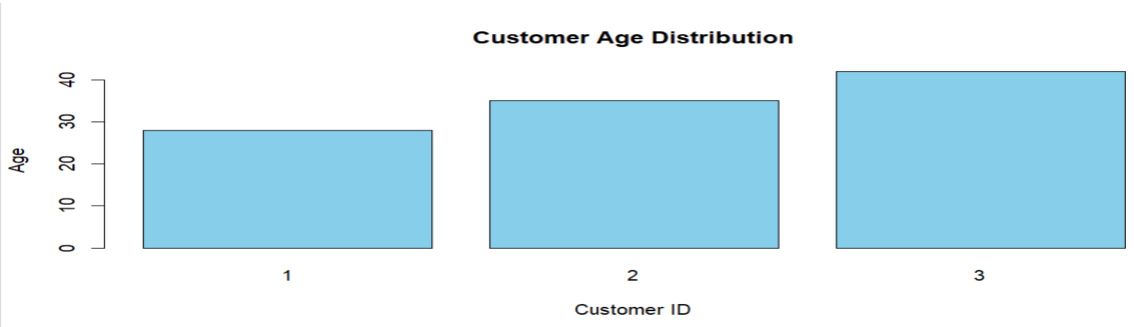
main="Gender Distribution")

3. Table

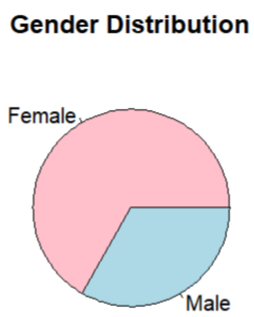
Customer

QUESTIONS

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to represent the distribution of customers by gender.



3. Build a table to show the demographic information for each customer. Label the table elements.

```
#-----------------------------
# 3. Table
#-----------------------------
customer
CustomerID Age Gender Income
         1  28 Female  50000
         2  35   Male  60000
         3  42 Female  75000
```

EXPERIMENT 8

Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

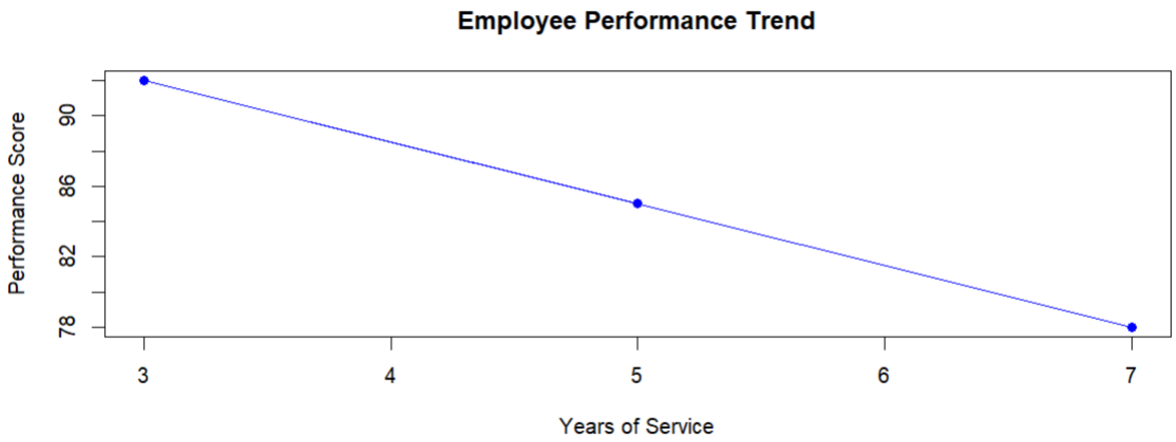
CODE:

```
# Employee Performance
employee <- data.frame(
 EmployeeID=c(1,2,3),
 Department=c("Sales","HR","Marketing"),
 Years=c(5,3,7),
 Performance=c(85,92,78)
)
# 1. Line Chart
plot(employee$Years,
   employee$Performance,
   type="o",
   pch=16,
   col="blue",
   xlab="Years of Service",
   ylab="Performance Score",
   main="Employee Performance Trend")
# 2. Department Bar Chart
```

```
dept <- table(employee$Department)
barplot(dept,
    col="orange",
    xlab="Department",
    ylab="Number of Employees",
    main="Employees by Department")
# 3. Table
Employee
```

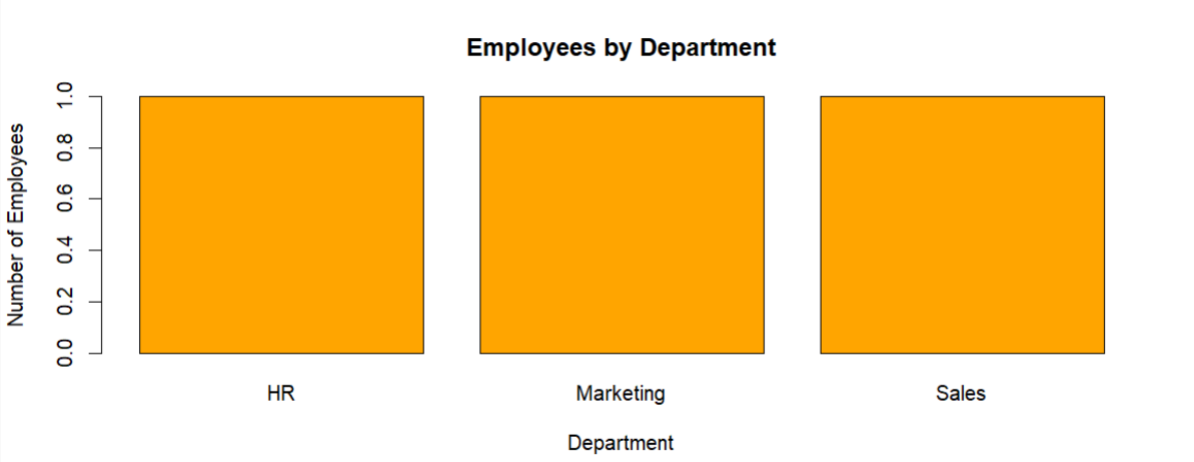
QUESTIONS:

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2. Generate a bar chart showing the distribution of employees across different departments.

Label the chart elements.



3. Build a table to display the performance data for each employee. Label the table elements.

```
#-----------------------------
# 3. Table
#-----------------------------
employee
EmployeeID Department Years Performance
         1      Sales     5          85
         2         HR     3          92
         3  Marketing     7          78
```

EXPERIMENT 9

Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in $)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

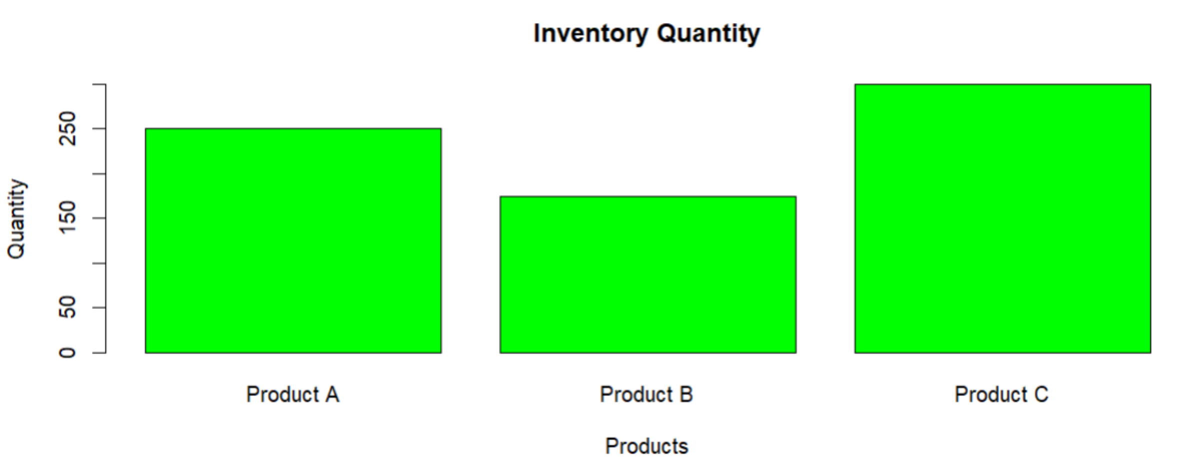
CODE

```
# Product Inventory
inventory <- data.frame(
 Product=c("Product A","Product B","Product C"),
 Quantity=c(250,175,300),
 Price=c(20,15,18)
)
# 1. Quantity Bar Chart
barplot(inventory$Quantity,
     names.arg=inventory$Product,
     col="green",
     xlab="Products",
     ylab="Quantity",
     main="Inventory Quantity")
# 2. Stacked Bar Chart
data <- rbind(inventory$Quantity,
          inventory$Price)
barplot(data,
```

```
        col=c("blue","red"),
        names.arg=inventory$Product,
        xlab="Products",
        ylab="Values",
        main="Quantity and Price")
legend("topright",
       legend=c("Quantity","Price"),
       fill=c("blue","red"))
# 3. Table
Inventory
```

QUESTIONS:

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

3. Build a table to show the inventory data for each product. Label the table elements.

```
#----------------------------
# 3. Table
#----------------------------
inventory
  Product Quantity Price
Product A      250    20
Product B      175    15
Product C      300    18
```

EXPERIMENT 10

Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

CODE:

Survey Responses

survey <- data.frame(

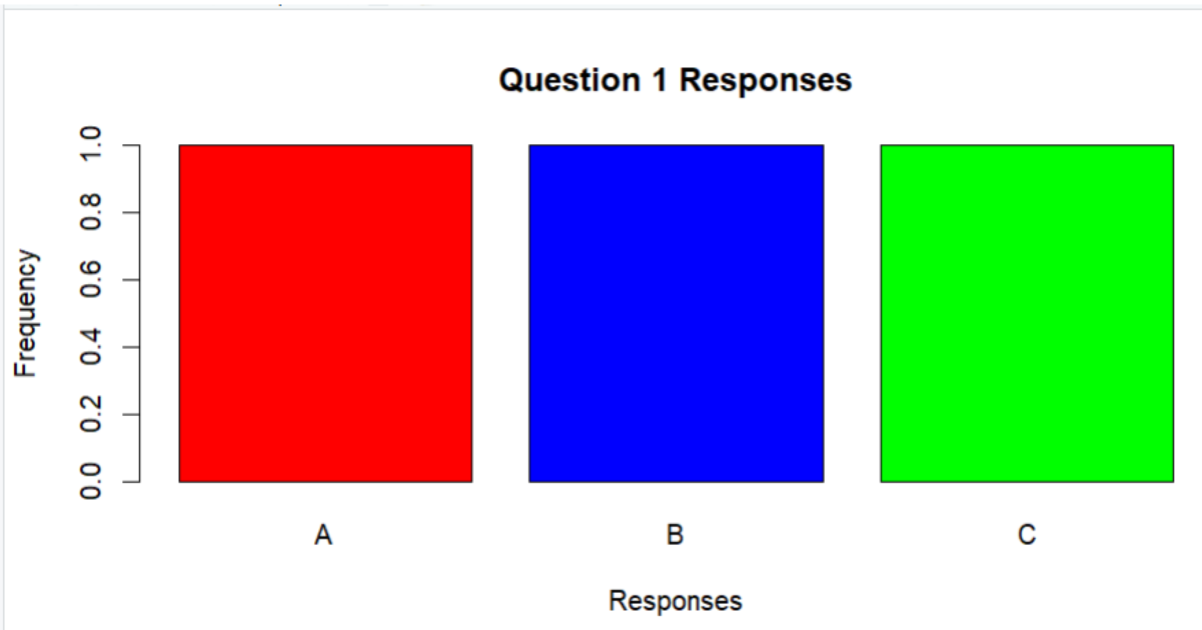
SurveyID=c(1,2,3),

Q1=c("A","B","C"),

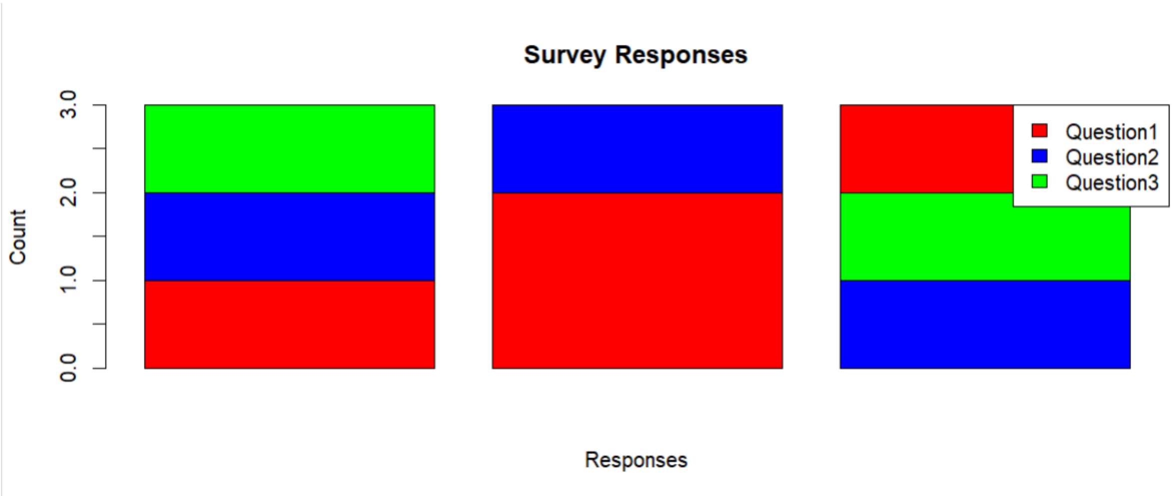
```
 Q2=c("B","A","A"),
 Q3=c("C","D","B")
)
# 1. Grouped Bar Chart (Q1)
q1 <- table(survey$Q1)
barplot(q1,
    col=c("red","blue","green"),
    xlab="Responses",
    ylab="Frequency",
    main="Question 1 Responses")
# 2. Stacked Bar Chart
responses <- rbind(
 table(factor(survey$Q1, levels=c("A","B","C","D"))),
 table(factor(survey$Q2, levels=c("A","B","C","D"))),
 table(factor(survey$Q3, levels=c("A","B","C","D")))
)
barplot(t(responses),
    beside=FALSE,
    col=c("red","blue","green"),
    xlab="Responses",
    ylab="Count",
    main="Survey Responses")
legend("topright",
   legend=c("Question1","Question2","Question3"),
   fill=c("red","blue","green"))
# 3. Table
survey
```

QUESTIONS:

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.



3. Build a table to show the survey response data for each survey. Label the table elements.

```
#----------------------------
# 3. Table
#----------------------------
survey
SurveyID Q1 Q2 Q3
       1  A  B  C
       2  B  A  D
       3  C  A  B
```